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I. Project Management Foundation

Q.1 Write short note on : Triple constraints.

⇒

- The triple constraint also known as Project Management triangle represents the three key factors that determine the project success.
  - 1) Time
  - 2) Scope
  - 3) Cost.

1) Time constraint

⇒ Time is very important project constraint.
- The amount of time required to complete a project or produce the deliverables must be estimated well for a good schedule.
- Projects often have deadlines to meet market needs or corporate goals.
- Failure to meet deadlines can lead to missed opportunities, increased expenses or loss of competitive advantages.
- Tools like Gantt charts and critical path analysis can help manage time constraints effectively.

2) Scope constraint

⇒ The scope of the project outlines the specific requirements or tasks which are necessary to complete the project.
- Managing the scope is vital in any project, whether they are agile

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software projects or well-planned waterfall projects.
- Failing to control the scope of the project will not help in delivering it on time or within budget.
- Any scope expansion ~~and~~ can increase costs and extends deadline.
- For example, If new feature added to a software project, it will require more time and budget.

3) Cost Constraint
⇒ It is the financial commitment that is made to the project.
- Cost estimating is used to figure out the financial commitment needed for all the resources necessary to complete the project.
- Cost budgeting is another process used to create a cost baseline.
- Cost control is third process which works to manage the Fluctuation of costs throughout the project.
- Cost has always been a complicated area on the Triple constraint triangle.
- To ensure that your estimates are accurate, it's advisable to use project management tools to calculate the cost variances.

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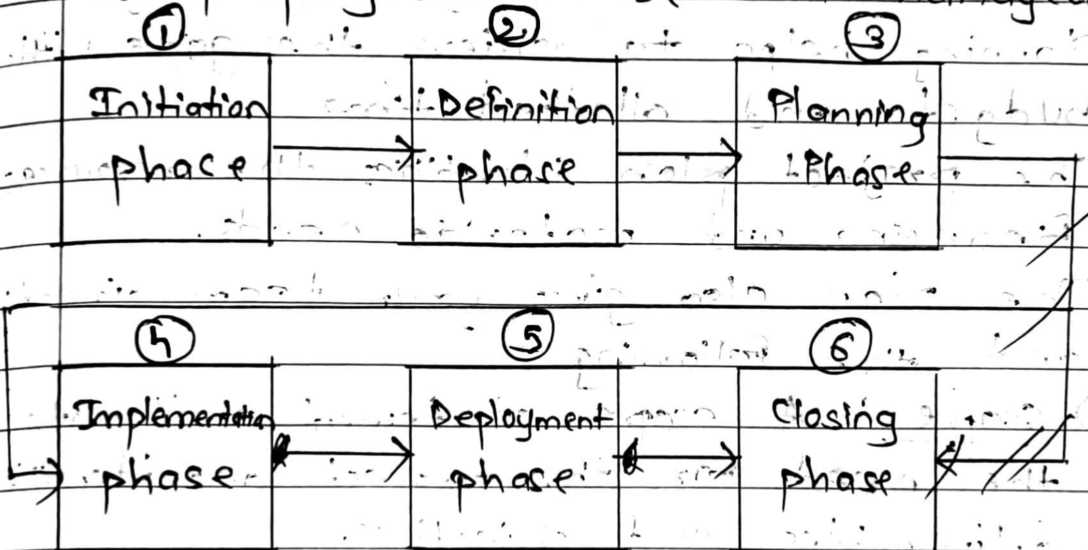
Q.2 What is project life cycle? How does cost of change, risk and influence of stakeholder are affected with project time during the life cycle of project?

⇒

- The project management life cycle is a process that is followed by nearly all the project managers.
- It provided a framework within which any project can be well managed.



1) Initiation Phase
⇒ This is the phase where all projects begins.
- In this phase the value as well as the feasibility of the project is determined.
- The project is approved or rejected on the basis of these two documents.
- They are created to convince the stakeholder or sponsors.

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2) <u>Defining phase</u>
⇒ In this phase, clearly define the project requirements and objectives.
- Establishing scope and getting detailed stakeholder requirements.

3) <u>Planning phase</u>
⇒ Once the project is approved, the next step is to build a project team and start working on project planning.
- Planning helps to achieve the goals within budget and allotted time.
- The project plan identifies the resources, financing and materials needs.
- The plan also gives your team direction and the following :-

a) <u>Scope</u> : A scope statement reiterates the need for the project and identifies deliverables and objectives.
b) <u>Definition</u> : It is a process used to break down the larger deliverables into smaller ones that help manage them better.
c) <u>Schedule</u> : This refers to the duration of the tasks and their completion data & time.
d) <u>Cost</u> : Costs are the financial commitments involved across the project and it helps formulate a budget.
e) <u>Tasks</u> : Tasks are performed to produce the deliverables.

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4) Implementation phase

⇒ After planning is done, it's time to start the project execution.

- This phase is made up of the following detailed processes:

a) Executing the Plan: You begin by following the plan you created. Assign the tasks to team members and manage and monitor their progress with project management tools.

b) Administrate: This is done by the managing the contract's secured in the project.

5) Deployment Phase

⇒ Delivering the final product or service to the client or end users.

- Transitioning from project work to ~~produce~~ operational use, including training and support.

6) Closing Phase

⇒ This is the last phase of the cycle.

- A project isn't over till the project goals and objectives have been met.

- The last phase of the project is all about meeting the goals.

- This involves the following set of processes:

a) Make sure the project deliverables have been completed as planned.

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4) Close out all outstanding contracts and administrative matters; archive the paperwork and disseminate to proper parties.

- Cost of change, Risk and Stakeholder Influence Over time:

1) Cost of change
⇒ The cost of change in projects refers to the budget, resources and time-related problems that arises when there are alterations to a project's specifications, scope or other aspect's after the project has begun.

2) Risk
⇒ The possibility of unexpected events affecting project goals. Risk is highest at the beginning and decreases as the project progresses and uncertainty reduces.

3) Stakeholder Influence
⇒ The ability of stakeholders to affect project decisions. Influence is strongest at the start and reduces as the project advances and ~~start~~ plans are finalized.

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Q.3 Explain project stage gate process of managing project life cycle. Explain how it helps top management in keeping project on right track

⇒

- This model offers a project management technique which a project is divided into several stages.
- This stages are separated by gates where decisions are taken to decide whether or not to proceed to the next stage.

1) Gates

⇒ Gates in ~~stage~~ state gate process are decisions points in a project.

- They help take a decision whether to continue the next stage or not.
- After each gate, one of the following decisions is to be taken.

a) Go: This decision considers the project good enough to move on to the next stage.

b) Kill: This decision is taken when the project is not good enough and cannot be developed further and terminating it is better.

c) Hold: This decision is taken project is not found good enough to continue at that moment, but there is possibility for the situation to improve

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and therefore rather than terminating it should be put on hold for resumption at a later date.

d) Recycle : This decision is taken when the project is found good enough to develop further with some changes.

2) Stages

⇒ The stage gate process has five stages.
- They are connected to each other by gates.
- Each stage is designed to collect specific information :

a) Stage 0 : Discovery

⇒ This is the initial preparatory stage. It identifies the project a company wants to undertake.
- For that, Ideas are generated in brainstorming sessions.
- Everyone from employees, customers to the suppliers are involved in the session.
- They provide useful information for idea generation.

b) Stage 1 : Scoping

⇒ In this stage, the product and the existing market for it are assessed.
- Product's strengths and weaknesses and the benefits it brings to the user are evaluated.

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- All the possible threats from competitors are also taken into account.

c) Stage 2: Business Plan Development
⇒ This is the last stage of concept development.
- Here, business plan is developed considering all kinds of opportunities, threats, competitions, etc.
- It is crucial before starting the actual project implementation.

d) Stage 3: Development
⇒ At this stage, the plans are executed and simple tests are conducted.
- For example, at this stage customers can be asked for their feedback of the product.
- A timeline with specific milestone that have to be achieved are created by the development team.

e) Stage 4: Testing and Validation
⇒ In this stage, product testing and validation are done.
- The manufacturing process and the product acceptance by customers and the market are assessed.

F) Stage 8: Launching and Implementation

⇒ In this stage, the marketing strategy comes into play.

- The product is ready to be launched and that requires attention by means of an advertising campaign, free publicity and interviews or other promotional activities.

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Q.4 Why negotiations are important in PM? Why win-win startergy is adopted in PM for negotiations?

⇒

- Negotiations is a process used to resolve disputes of different kinds between different people or group.
- Negotiation involves a discussion between two or more parties involved.
- It aims at reaching an agreement or settlement.
- It can take place at any time in a project, program or portfolio and it may be formal or informal in nature.
- Importance :

1) Conflict resolution

⇒ Helps solve disagreements between stakeholders or team members.

2) Resource allocation

⇒ Ensures optimal sharing of limited resources like manpower, budget and equipment.

3) Scope management

⇒ Negotiates changes to scope, timelines or cost without harming project goals.

4) Building relationships

⇒ Creates trust and cooperation among all involved parties.

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- **Win-win stratergy** :

- The win-win stratergy aims for solutions where all parties benefit, ensuring satisfaction for everyone involved rather than one side "winning" and the other "loosing".

1) **Long-term relationships**
⇒ PMs often work with the same vendors, clients or teams on multiple projects. Win-win builds strong, lasting partnerships,

2) **Better collaborations**
⇒ Satisfied stakeholders and teams are more motivated to contribute fully.

3) **Higher project success rate**
⇒ Projects succeed more often when everyone is committed and feels hard.

4) **Reduced resistance**
⇒ When both sides feel respected, they are less likely to resist project changes or decisions.

5) **Trust building**
⇒ Win-win creates an atmosphere of fairness and trust, making future negotiations smoother.

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Q.3 Differentiate between the functional, Pure project and matrix organization.

⇒

Functional	Pure Project	Matrix
1) Functional structure divides the organization based on specialized functional areas such as production, sales and marketing.	1) Project organization is a separate form of functional unit in which all members of project work force are directly responsible to project manager.	1) Matrix structure is a type of organization structures where employees are grouped concurrently by two different operational dimensions.
2) Functional structure is simple and convenient to manage.	2) Projectized organizations are organized around projects for maximal PM effectiveness.	2) Matrix structure is complex in nature due to the combination of two organizational structures
3) Functional structure is suitable for organizations that operate in a single location with single product category	3) Project structure is especially suitable for ~~organizations that operate in a single locatio~~ complex, innovative environments	3) Matrix structure is appropriate for companies that have multiple product categories and carries out various projects.

Functional	Pure project	Matrix
4) In this, there is no one on the project team.	4) In this, essentially everybody is on the project team.	4) The matrix falls in between and includes a variety of organizational alternatives ranging from a weak to strong matrix.
5) Functional manager has full control.	5) Project manager has little authority.	3) Authority is shared between functional and project manager.
6) Project focus is low.	6) Project focus in high.	6) Project focus is moderate.
7) Employees have permanent role in the department	7) Employees may need to find new project	7) Employees can return to their functional roles after projects

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Q.6 Why project manager's role is more of facilitator rather than a supervisor?

=>

- A project manager is responsible for leading a project from initiation to completion, ensuring that project meets its objectives.
- A project manager's role is often more of a facilitator than a supervisor for several key reasons:-

1) Collaboration over authority:

=> The project manager ensures that all team members from different functional areas collaborate effectively, remove obstacles and help ~~them~~ team stay aligned with the project goals.

- In a traditional supervisor role, the focus would be more on directing and controlling every aspect of team member work.

2) Focus on team dynamics

=> A project manager helps foster a collaborative team environment, ensuring that communication flows smoothly and conflicts are resolved.

- Supervisors often focus more on managing individual performance and ensuring adherence to specific tasks rather than guiding team collaboration.

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3) Adaptability

⇒ A project manager must be flexible and adapt to changing circumstances in a project.

- Supervisors may be more rigid, sticking to predefined processes and structures, with less focus on adapting to new situations outside of their specific scope.

4) Goal-oriented leadership

⇒ Project managers are generally focused on achieving project goals through teamwork, coordination and effective management of time, scope and cost.

- Supervisors tend to focus more on overseeing specific tasks or the performance of individual team members within their functional areas, not necessarily the broader project goal.

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Q.7 Write short note on : ~~Project~~ Role of Project manager.

=>

- A project manager is responsible for leading a project from initiation to completion, ensuring that project meets its objectives.

- Key Roles and Responsibilities :

1) He designs the appropriate project management standards and then implements that.
2) He also has to manage the production of the required deliverables at different stages of the Project development.
3) He has a crucial role of preparing projects plan and ways to monitor the project.
4) He also prepares and maintains project stages and exception plans as and when required.
5) He has to identify and managing project risks including the development of contigency plans.
6) He has to reporting to the top management on project progress through reports and end stage assessment.
7) He monitors the overall progress and usage of resources.

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8) Tracks the actual progress and compares with the planned progress

9) Keeps stakeholder Informed about the progress.

10) He has ~~to~~ regularly conducting a project evaluation review to assess how well the project is being managed. ~~and~~

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Q.8 Who are the stakeholders in project? Why communication is most important ~~layer~~ part of the project manager's job?

⇒

- **Stakeholders :**

- Stakeholders are individuals, groups, or organizations who have an interest in the outcome of the project.
- They can influence or be affected by the project's execution or results.
- Common project stakeholders :

1) Project sponsor
⇒ Provides financial support and overall direction.

2) Clients / customers
⇒ End-users or organizations who will use the project deliverables.

3) Project team members
⇒ Individual doing the project work.

4) Functional managers
⇒ Managers who control resources or provide expertise.

5) Suppliers
⇒ External parties providing goods or services to the project.

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- Why communication is important?

- Communication ensures that stakeholders and team members understand the project goals, requirement and changes clearly.
- Regular, transparent communication builds trust among stakeholders and reduces uncertainty.
- Good communication prevents misunderstanding, misinterpretations and mistakes in project execution.
- Quick and accurate communication enables faster decision-making and problem-solving.
- Proper communication helps in setting and managing realistic expectations among clients, sponsors and team members.
- Open communication channels help address issue early, avoiding major conflicts later.
- Through status updates and ~~monitoring~~ reporting, the project manager ensures everyone is aware of the project's progress, risks and next steps.

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Q.9 What are the attributes of good project manager? explain with suitable example.

⇒

- A good project manager possesses a mix of technical skills, leadership qualities and personal traits that help guide a project to success.
- Attributes :-

1) Leadership
⇒ Ability to inspire, motivate and guide the team towards achieving project goals.
- For e.g.,
A PM encourages a team under tight deadlines to stay positive and focused, leading by example.

2) Communication Skills
⇒ Ability to clearly share information with stakeholders and team members.
- For e.g.,
A PM regularly updates the client and team through meetings and reports, preventing misunderstandings.

3) Problem-solving ability
⇒ Skill in identifying problems quickly and finding effective solutions.
- For e.g.,
When a key supplier fails to deliver, the PM immediate finds an alternate supplier to avoid project delays.

4) Time Management
⇒ Managing time efficiently to ensure the project stays on schedule.
- For e.g.,
A PM prioritizes critical tasks and delegates minor ones to meet the final deadlines.

5) Risk Management
⇒ Proactively identifying risks and planning ways to handle them.
- For e.g.,
A PM anticipates potential delays due to whether in a construction project and creates a backup plan.

6) Decision-making ability
⇒ Making quick and effective decisions, especially under pressure.
- For e.g.,
A PM selects a more reliable vendor at the last moment when the original vendor backs out.

7) Adaptability
⇒ Adapting plans and strategies when project conditions change.
- For e.g.,
When the client's requirements shift mid-project, the PM reworks the schedule and scope without panic.

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Q.10 Describe the PM knowledge areas as per PMI in brief.

⇒

- The project management knowledge areas are essentially what you need to know about effective project management.
- PM knowledge areas as per PMI :-
  - 1) Integration
  - 2) Scope
  - 3) Time
  - 4) Cost
  - 5) Quality
  - 6) Procurement
  - 7) HR
  - 8) Communications
  - 9) Risks
  - 10) Stakeholders

1) Integration

⇒ Project integration management is the umbrella that covers all other project management knowledge areas.

- This knowledge area focuses on ensuring that all parts of the project work together seamlessly.
- It involves developing the project charter, creating a project management plan, directing and managing project work and ensuring that project is closed properly.

2) Scope

⇒ It is the way to define what your project will deliver.

- Scope management is all about making sure that everyone is clear about what the project is for and what it includes.
- It covers collecting requirements and preparing the work breakdown structure.

3) Time

⇒ It relates to how you manage the time, people are spending on their project tasks, and how long the project takes overall.

- This knowledge areas help you understand the activities in the project, the sequence of those activities and how long they are going to take.

4) Cost

⇒ Cost management is about planning and controlling the project's budget.

- It involves estimating costs, determining a budget, and controlling actual expenditures to ensure the project stays within its financial constraints.
- Proper cost management helps avoid budget overruns.

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5) Quality

⇒ This knowledge areas ensures that the project will satisfy the needs for which it was undertaken.

- It involves planning for quality, performing quality assurance activities, and controlling quality through inspections and testing.
- Delivering a product or service that meets the quality standard is essential for customer satisfaction.

6) Procurement

⇒ This knowledge area deals with obtaining goods and services from outside the project team.

- It includes planning procurements, conducting procurements, controlling contracts and closing procurements.
- Managing procurement properly ensures that all outside resources contribute effectively to the project.

7) HR

⇒ Human resources are vital to complete your project so you need to put your team together.

- After that, it's all about managing the people on the team including giving them extra skills to do their jobs.

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8) Communications

⇒ Effective communication is critical to the project success.

- This area involves planning communications, managing and monitoring communications to ensure that project information is properly generated, collected, distributed and stored.
- Good communication keeps stakeholders informed and engaged.

9) Risk

⇒ Risk management focuses on identifying, analyzing and responding to project risks.

- It includes planning for risk management, identifying potential risks, performing risk analysis and monitoring risks throughout the project.
- Proactively managing risks minimizes their impact on the project.

10) Stakeholder

⇒ Stakeholder management involves identifying all people or organizations affected by the project, understanding their expectations and influence, and developing stratergies to engage them appropriately.

- It ensures that stakeholder's needs are met and that their support is is maintained throughout the project life cycle.

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Q.11 Discuss why project management is essential in today's business environment. What benefits does it provide in achieving organizational goals?

⇒

- In today's fast-moving and competitive business world, companies deal with tight deadlines, limited budget, complex tasks and high customer expectation.
- Without proper planning and coordination, things can easily go off track.
- This is where project management becomes very important.
- Project management helps organize, plan, execute and deliver projects successfully making sure everything runs smoothly, on time and within budget.
- Key Reasons

⇒ 1) Clear planning and direction
2) Better use of Time and Resources
3) Risk Management
4) Improved Communication
5) Quality Control
6) Better decision making
7) Achieving business goals
8) Customer satisfaction.

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Q.1 Define and differentiate projects and operations.

⇒

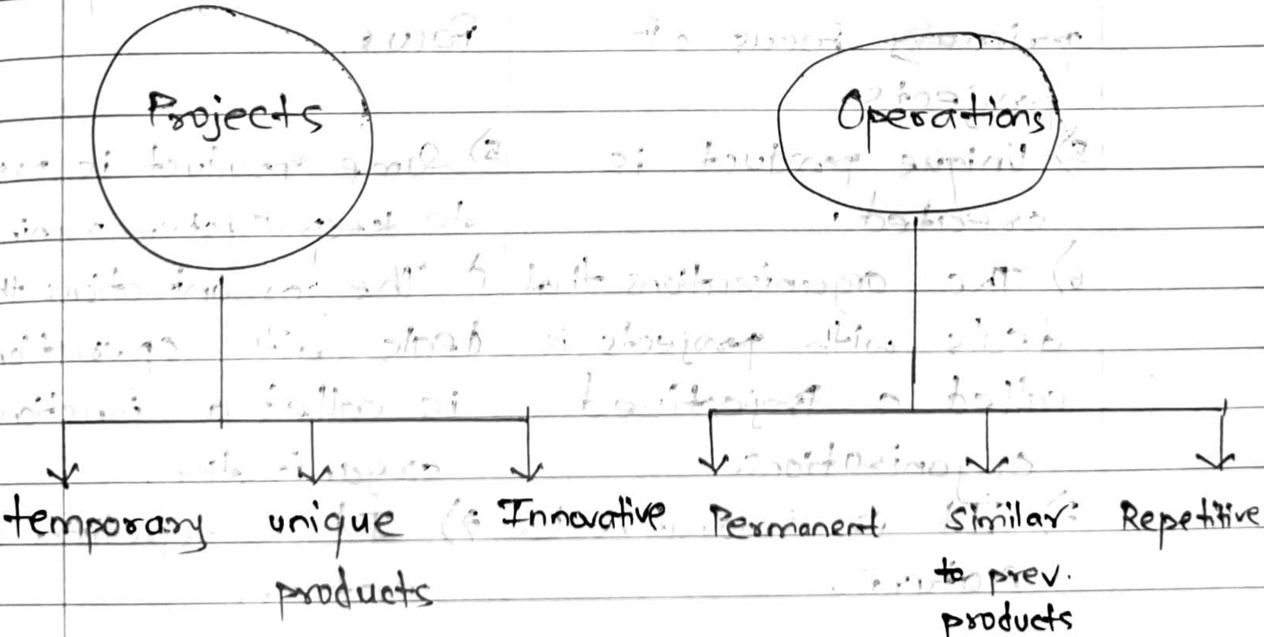
1) Project

⇒ A project is a temporary work done with a define start and end.

- It is generally unique in nature unlike a regular operation.

2) Operations

⇒ The processes of creating product again and again using same techniques that were used earlier, for purpose of distributing to users and are a part of Operations.



Projects	Operations
1) A project is an endeavor that is temporary in nature, that is undertaken to produce a unique product.	1) Operations are ongoing execution of activities which occurs after product is made to produce same result or a repetitive service.
2) The budget is defined for projects.	2) The budget is not defined for Operations as the earning needs to be done to keep operations alive.
3) It has more risk as it is done for first time.	3) It has less risk as such products have already been made before.
4) Performance is primary focus of projects.	4) Efficiency is primary focus.
5) Unique product is created.	5) Same product is produced to keep system running.
6) The organizations that deals with projects is called a Projectized organizations.	6) The organizations that deals with operations is called a functional organization.
7) It is innovative in nature.	7) It is repetitive in nature.